Care Transition Efficiency & Placement Outcome Analytics: A Process Flow & Little's Law Evaluation of the UAC Pipeline

Unified Mentor Research Initiative | In Collaboration with HHS & CBP Operational Datasets

Abstract: *The Unaccompanied Alien Children (UAC) care pipeline represents a high-stakes, multi-agency humanitarian and child welfare network. Historically, federal oversight has focused on static headcounts rather than dynamic process metrics, obscuring velocity bottlenecks and queue delays. This study formulates a rigorous flow pipeline model tracking daily transitions across 720 reporting dates from 2023 to 2025. Applying Little's Law from queueing theory and deriving novel transition metrics—including the Transfer Efficiency Ratio, Discharge Effectiveness Index, and Outcome Stability Score—we uncover critical system mechanics. CBP facilities demonstrated an average residence time of 1.33 days (86.3% TVPRA compliance), while HHS shelters maintained an average length of stay of 34.9 days. Administrative batching on weekends was identified as a persistent operational bottleneck. We conclude with data-driven policy recommendations to stabilize transitions and accelerate safe sponsor reunifications.*

1. Introduction & Statutory Framework

The United States child welfare and immigration architecture governing unaccompanied minors is anchored in the William Wilberforce Trafficking Victims Protection Reauthorization Act of 2008 (TVPRA, 8 U.S.C. § 1232) and the Flores Settlement Agreement. TVPRA mandates that, except in exceptional circumstances, unaccompanied children encountered by Customs and Border Protection (CBP) must be transferred to the custody of the Department of Health and Human Services (HHS), specifically the Office of Refugee Resettlement (ORR), within 72 hours of apprehension. ORR is responsible for housing minors in the least restrictive setting while conducting thorough background vetting to reunify them with vetted sponsors.

2. Mathematical Formulation of Process Metrics

To evaluate transition velocity, we introduce five standardized process metrics grounded in continuous-time queueing theory:

$$\text{Transfer Efficiency Ratio} = \text{Transfers}_t \,/\, \text{CBP_Custody}_t$$

$$\text{Discharge Effectiveness} = \text{Discharges}_t \,/\, \text{HHS_Care}_t$$

$$\text{Pipeline Throughput Rate} = \text{Discharges}_t \,/\, \text{Apprehensions}_t$$

$$\text{Little's Law Residence Time: } W = L \,/\, \lambda \rightarrow \text{Stay_Days} = \text{Census} \,/\, \text{Daily_Exit_Rate}$$

$$\text{Outcome Stability Score} = 100 \,/\, (1 + \text{CV_30d(Discharges)})$$

3. Empirical Findings & Longitudinal Analysis

Analyzing 720 daily government reports reveals three distinct operational phases: (1) High-Volume Equilibrium in 2023 (daily mean intake: 117.6; discharges: 288.0), (2) Surge & Capacity Stress in 2024 (daily intake peak: 225.8 in Feb 2024; active HHS care peaking over 11,000 children), and (3) Policy-Induced Low-Volume Retraction in 2025 (mean daily intake: 13.0; shelter census under 1,000). Cross-correlation analysis indicates that intake shocks propagate to discharge volumes with an empirical lag of 14 to 21 days, reflecting the required sponsor screening and medical clearance lifecycle.

4. Policy & Operational Reforms

We propose three concrete structural reforms: First, institutionalize continuous 7-day case management to eliminate the 32% weekend discharge deceleration. Second, integrate live algorithmic bottleneck alerting into the ORR bed placement engine. Third, establish regional foster family surge reserves to prevent reliance on high-cost influx facilities during sudden migration shifts.